

DIURETICS

The diuretics are the drugs which primarily cause a net loss of Na^+ (*natriuresis*) associated with water loss (secondary to natriuresis) and thus increase the rate of urine flow.

Classification

Depending upon their efficacy, the diuretic drugs can be classified as:

1. **High-efficacy diuretics** (inhibitors of Na^+ - K^+ - 2Cl^- transport), also called loop diuretics, e.g. Furosemide.
2. **Medium efficacy diuretics** (inhibitors of Na^+ - Cl^- symport)
 - (i) Thiazide diuretics, e.g. chlorothiazide.
3. **Weak or adjunctive diuretics**
 - (i) Carbonic anhydrase inhibitors, e.g. acetazolamide
 - (ii) Potassium sparing diuretics, e.g. spironolactone, triamterene and amiloride
 - (iii) Osmotic diuretics, e.g. mannitol, isosorbide and glycerol
 - (iv) Xanthines, e.g. theophylline.

Site of action, mechanism of action and major effects of various classes of diuretics

Table 6.6-2 Site of action, mechanism of action and major effects of various classes of diuretics

Class of diuretic	Site of action	Mechanism of action	Major effects
High efficacy diuretics Loop diuretics, e.g. – Furosemide – Bumetanide – Ethacrynic acid	TAL	Inhibition of $\text{Na}^+-\text{K}^+-2\text{Cl}^-$	$\uparrow$ NaCl excretion $\uparrow$ K^+ excretion $\uparrow$ Ca^{2+} excretion $\downarrow$ Ability to concentrate urine $\downarrow$ Ability to dilute urine
Medium efficacy diuretics Thiazide diuretics, e.g. – Chlorothiazide – Hydrochlorothiazide	Early distal tubule	Inhibition of Na^+-Cl^- symport	$\uparrow$ NaCl excretion $\uparrow$ K^+ excretion $\downarrow$ Ca^{2+} excretion $\downarrow$ Ability to dilute urine
Weak or adjunctive diuretics (i) Carbonic anhydrase inhibitor, e.g. – Acetazolamide (ii) Potassium sparing. diuretics, e.g. – Spironolactone – Triamterene – Amiloride (iii) Osmotic diuretics, e.g. – Mannitol – Isosorbide	Proximal tubule Late distal tubule and collecting duct Proximal tubule	Inhibition of carbonic anhydrase Inhibition of Na^+ reabsorption, inhibition of K^+ secretion Retain water iso-osmotically and inhibits NaCl reabsorption	$\uparrow$ HCO_3^- excretion $\uparrow$ Na^+ excretion $\downarrow$ K^+ excretion $\downarrow$ H^+ excretion $\uparrow$ Na^+ excretion $\uparrow$ K^+ excretion $\uparrow$ Cl^- excretion

Class of diuretic	Site of action	Mechanism of action	Major effects
High efficacy diuretics Loop diuretics, e.g. • Furosemide • Bumetanide • Ethacrynic acid	Thick ALOH (TAL)	Inhibition of $\text{Na}^+-\text{K}^+-2\text{Cl}^-$	$\uparrow$ NaCl excretion $\uparrow$ K^+ excretion $\uparrow$ Ca^{2+} excretion $\downarrow$ Ability to concentrate urine $\downarrow$ Ability to dilute urine
Medium efficacy diuretics Thiazide diuretics, e.g. • Chlorothiazide • Hydrochlorothiazide	Early distal tubule	Inhibition of Na^+-Cl^- symport	$\uparrow$ NaCl excretion $\uparrow$ K^+ excretion $\downarrow$ Ca^{2+} excretion

Class of diuretic	Site of action	Mechanism of action	Major effects
			↓ Ability to dilute urine
<i>Weak or adjunctive diuretics</i> (i) Carbonic anhydrase inhibitor, e.g. - Acetazolamide (ii) Potassium sparing diuretics, e.g. - Spironolactone - Triamterene - Amiloride (iii) Osmotic diuretics, e.g. - Mannitol - Isosorbide	Proximal tubule Late distal tubule and collecting duct Proximal tubule	Inhibition of carbonic anhydrase Inhibition of Na^+ reabsorption, inhibition of K^+ secretion Retain water iso-osmotically and inhibits NaCl reabsorption	↑ HCO_3^- excretion ↑ Na^+ excretion ↓ K^+ excretion ↓ H^+ excretion ↑ Na^+ excretion ↑ K^+ excretion ↑ Cl^- excretion

Other substances which act as diuretics

- Water*. It acts by inhibiting antidiuretic hormone (ADH) secretion
- Alcohol*. It also acts by inhibiting ADH secretion
- Antagonist of V_2 vasopressin inhibitors* inhibit action of vasopressin on the collecting ducts
- Glucose* (e.g. in diabetes mellitus). It acts as an osmotic diuretic
- Caffeine*. It increases GFR and probably decreases renal tubular Na^+ reabsorption.